

XSFN-Net: A Hybrid Deep Spiking Fractional Neural Network for OSCC Classification in Histopathological Images

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Abstract—Oral Squamous Cell Carcinoma (OSCC) is one of the most Prevalent forms of oral cancer, commonly associated to risk factors like tobacco use, alcohol use, chewing betelnut and viral infections. Early diagnosis and treatment of OSCC and other oral mucosal potentially malignant lesions is the clinician's best bet to improve prognosis .In this research paper, we propose the new Deep Learning model of Xception spiking Fractional Neural Network (XSFN-NET) to different OSCC of the histopathology. The process starts with the image intensification with the Medav filter, then accurate pixel-wise segmentation with the Parallel Reverse Attention Network. Feature extraction is done with a combination of hand-crafted methods—Local Vector Pattern (LVP), Discrete wavelet transform (DWT), and Gray-level run-length matrix (GLRM)—and the deep features from the Xception model. The features are the combined using Fractional Calculus (FC) and classified by a Deep Spiking Neural Network (DSNN) to make efficient and biologically inspired decisions. The result of XSFN-Net reached an accuracy of 91%. Its precision was 93.51%, recall was 93.01%, and the F1-score was 93.64%.

Index Terms—OSCC, Deep Learning, Spiking Neural Networks, Histopathology, Fractional Calculus, Xception

I. INTRODUCTION

Approximately 84–97% of malignancies of the oral cavity is the oral squamous cell carcinomas, is the common types of the oral cancer. It starts in the squamous cells that lines oral cavity and is the frequently connected to tobacco use and other lifestyle choices. use, drinking alcohol, chewing betel nuts, and being infected with viruses like HPV.

Early detection of OSCC is crucial because it significantly boosts survival rates. Patients diagnosed early have an 85% five-year survival rate, while those diagnosed at a later stage only have a 40% five-year survival rate. Early detection of OSCC is crucial, as it significantly improves survival rates. If identified in the early stages, patients have a five-year survival rate exceeding 80%, but this drops to below 50% in advanced stages.

Despite advances in cancer research and molecular biology, early detection of OSCC remains extremely challenging. Most

patients are diagnosed later in life because early lesions are often subtle and asymptomatic, which lowers survival rates and creates a negative outlook. To detect signs of OSCC, physicians use scans, physical examinations, and reviews of symptoms alongside tissues examinations through biopsies.

Traditional methods face challenges microscopy and staining as well as different levels of experience and artistry of different pathologists, making them tedious and imprecise. This subjectivity is tied to the risks of inaccurate diagnosis and the risks of “anesthesia of the awakened subject” [1].

For OSCC, features are image-based and nonimage-based. They are all blended and processed with a CNN. This approach has the potential to develop low-cost AI algorithms for cancer screening a AI-based algorithms for screening and diagnosis of cancer in underdeveloped and poor regions [2]. This work proposes an ensemble architecture for OSCC classification from histopathological images using the Inception ResNetV2 and Xception models. Using both networks in the ensemble approach improves generalization and feature diversity, two benefits of ensemble approaches that increase accuracy [3].

The model in [5] was designed to simplify deployment for primary care clinics, field healthcare settings, and rural screening camps since it runs on low-power devices. This research introduces a hybrid ensemble model which features a unique Convolutional VGG. The model implements explainable artificial intelligence (XAI) through LIME to boost both prediction accuracy and result interpretability [8]. The model employs gradient-boosted decision trees together with deep feature extraction to merge data-driven learning with structured decision-making. The ensemble models built from EfficientNet architectures demonstrate superior OSCC detection accuracy while maintaining the lowest computational requirements according to [10].

II. LITERATURE STUDY

Ramya et al. [1] created XSFN-Net which functions as a hybrid framework by uniting Deep Spiking Neural Networks (DSNN) with handcrafted features consisting of texture-based Local Vector Patterns (LVP), frequency-based Discrete-WaveletTransform (DWT) and statistical descriptors using GrayLevelRunMatrix (GLRM). The framework also integrates pixel-level lesion segmentation using PraNet. G.A.I. Devindi et al. [2] developed a Multimodal deep convolutional neural network pipeline aimed at the early detection of the oral cancer. Their model utilizes images of the oral cavity captured via smartphone, alongside patient information such as age, gender, and lifestyle habit to enhance prediction accuracy. While the early-fusion multimodal process achieved only 81%, it reflects genuine clinical practice workflows, leading to a more complete and accessible diagnostic model which is more practical in a healthcare setting.

Madhusmita Das et al. [3] developed an ensemble deep learning model for OSCC detection, employing a two-stage technique to select and combine the best-performing pre-trained models—InceptionResNetV2 and Xception. This architecture capitalizes on ensemble learning's strength in improving prediction performance, even without relying on hand-crafted features or custom segmentation. Researchers from Amity University [4] focused on developing a lightweight OSCC detection system using CNN and TensorFlow Lite. Although the paper lacks comprehensive performance metrics, it demonstrates progress toward creating mobile and edge-compatible solutions for low-cost, real-time screening in remote or resource-constrained areas.

S. Vasuki et al. [5] proposed an ensemble deep learning framework that combines modified VGG networks and CNNs for feature extraction, followed by classification using XG-Boost. This hybrid approach effectively captures nonlinear feature relationships, improving both classification accuracy and model generalization. Desai et al. [6] introduced a lightweight diagnostic model using DenseNet201 and FixCaps applied to mobile-captured oral cavity images. Their work focuses on early screening of OSCC and potentially malignant disorders, offering a portable and effective solution for community-level diagnostics. Somyanonthanakul et al. [7] extended the research scope by introducing a Fuzzy Deep Learning (FDL) model that incorporates both clinical and pathological data to predict patient survival outcomes.

Samiha Islam et al. [8] underscored the need for explainability for clinical AI applications with the combination of EfficientNetB3, along with LIME (Local Interpretable Model-agnostic Explanations) which provided visual rationalization for model predictions. B. Ananthakrishnan et al. [9] proposed a Deep Neural Network-based pipeline for the automatic detection and classification of the OSCC with histopathological images; also achieved high sensitivity and specificity on test sets.

Albalawi et al. [10] validated the capabilities of EfficientNet architectures for OSCC detection using standard histopatho-

logical data sets. Their study revealed both high classification and classification robustness, confirming the model's efficacy for real-world utilization.

III. PROPOSED METHODOLOGY

A. Dataset Description

Dataset-1 comprises 1,224 histopathological images utilized for the classification of Oral Squamous Cell Carcinoma (OSCC). The images are collected at two different Magnification levels. At 100x Magnification, there are 89 images of normal oral Epithelium and 439 images of OSCC tissue. At 400x Magnification, the dataset includes 201 images of normal Epithelium and 495 images of OSCC. These images offer a diverse representation of both healthy and cancerous tissue, aiding in effective model training and evaluation for automated OSCC detection.

B. Methods of Preprocessing

Before the classification model is trained, the input images go through a number of meticulously planned preprocessing steps to standardize, improve, and augment them. Increasing dataset diversity, normalizing spatial scales, and lowering noise are the goals of these operations.

Table I: Preprocessing Techniques

Step	Technique	Purpose
1	Medav Filter	It removes noise and preserves fine tissue structure
2	Parallel Partial Decoder	Aggregates high-level semantic features.
3	Reverse Attention Module	Refines coarse segmentation maps by focusing on boundary regions.
4	Loss Function	Combines Tversky Index and weighted binary cross-entropy to balance false positives and false negatives.
5	Local Vector Pattern	Captures textural and structural orientation
6	Discrete Wavelet Transform	Encodes spatial-frequency information across scales.
7	Gray level run-length matrix	Extracts statistical texture features like run-length and gray-level variations.

1) *Image Enhanced*: All images are initially passed through the Medav Filter for preprocessing. This filter is a hybrid of adaptive median and mean filters, designed to:

- Remove impulse noise,
- Enhance tissue contrast,
- Preserve fine histological details.

This step improves the input quality for accurate segmentation and feature extraction.

2) *Pixel-Wise Segmentations*: The enhanced images are segmented using the Parallel Reverse Attention Network (PraNet).

Key components include:

- **Parallel Partial Decoder**: Aggregates high-level semantic features.
- **Reverse Attention Module**: Refines coarse segmentation maps by focusing on boundary regions.

- **Loss Function:** Combines Tversky Index and weighted binary cross-entropy to balance false positives and false negatives.

The segmentation output highlights the pathological regions (potential OSCC zones) at the pixel level.

3) *Feature Extractions:* From the segmented images, relevant features are extracted using:

- 1) **Local Vector Pattern (LVP):** Captures textural and structural orientation.
- 2) **Discrete Wavelet Transform (DWT):** Encodes spatial-frequency information across scales.
- 3) **Gray Level Run-Length Matrix (GLRM):** Extracts statistical texture descriptors:
 - Run-Length Non-Uniformity (RLN)
 - Gray-Level Non-Uniformity (GLN)
 - Long Runs
 - Short Runs

The final feature vector is composed of these six extracted features.

The Flow chart for the Feature Extraction by using the above techniques can be shown in the below.

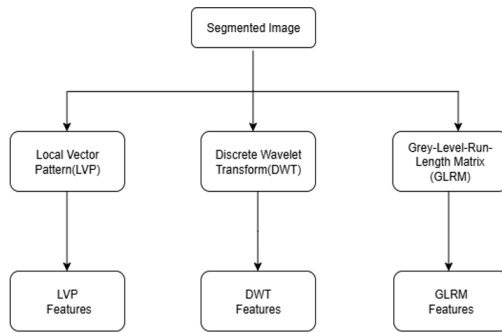


Figure 1: Feature Extraction.

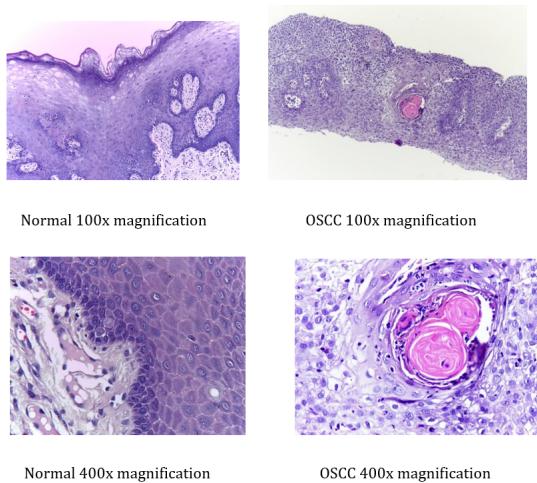


Figure 2: Before Preprocessing.

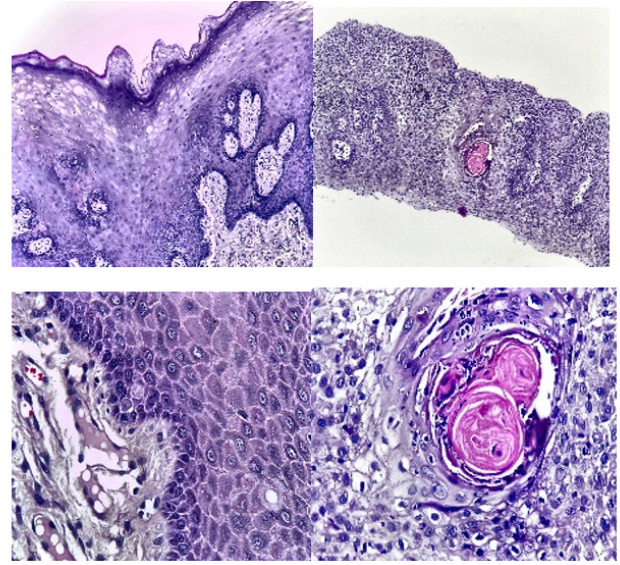


Figure 3: After Preprocessing.

C. General Process Flow

The filtered image is passed into the XSFN-Net Computation block, which consists of a series of four convolutional layers. These convolutional layers are responsible for extracting deep spatial features that capture the intricate patterns and textures unique to cancerous and non-cancerous tissues. To reduce the dimensionality and retain the most prominent features, the model employs an Average Pooling operation with a 3×3 convolution, which also aids in reducing computational complexity and mitigating overfitting.

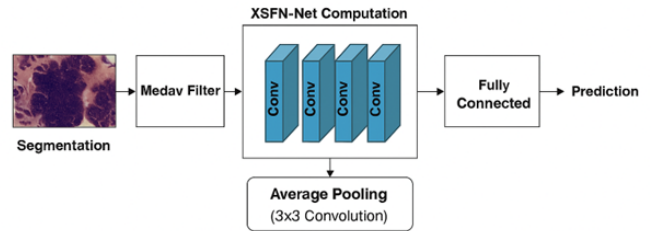


Figure 4: Block Diagram of the Proposed XSFN-NET Architecture

The features are subsequently sent over to a Fully Connected Layer, which transforms the features collected, into a one-dimensional vector for classification. In this stage, we will likely see a dropout regularization layer in order to enhance generalizability. Ultimately, the training model produces a

prediction through the prediction layer, based on the learned features, to determine if the image is Normal or OSCC. This end-to-end approach relies on automatic preprocessing and deep feature extraction to provide a fast and accurate solution for OSCC detection.

D. Model Architecture

Xception Spiking Fractional Neural Network, is the hybrid Deep Learning architecture designed for the accurate classification of the Oral Squamous Cell Carcinoma (OSCC) from histopathological images. This model combines the strengths of three powerful concepts: the Xception deep convolutional network, fractional calculus based feature fusion and deep spiking neural networks (DSNNs).

The XSFN-Net model operates as follows: the Medav Filter is first applied to input images - this is a new preprocessing filter that combines the adaptive median filter and means filter for noise reduction and tissue structures preservation. Subsequently the image is segmented into lesion areas (which has typically achieved using PraNet), and then features are extracted from the image in two streams, and attempted parallelism: handcrafted features (DWT, LVP. and GLRM), and deep features.

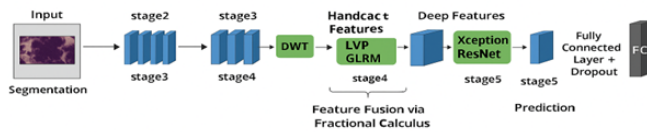


Figure 5: Architecture of the OSCC with Fractional Calculus Feature Fusion

The combination of deep and handcrafted features is done through a fractional calculus-based approach, which gives an edge in comparison to traditional approaches, providing a richer and flexible information integration. This fusion layer effectively combines handcrafted textural patterns with deep spatial features. The fused representation is passed through a fully connected layer with dropout for dimensionality reduction and regularization. Finally, the classifier outputs the prediction as either Normal or OSCC based on the learned patterns.

The calculated architecture is attempting to leverage the best of both domain specific handcrafted features and deep representations, to create a potentially more powerful and effective model for OSCC detection.

The Xception architecture is built upon depthwise separable convolutions, which extract high-level semantic aspects of an

image to allow capturing of granular morphologic distinctions between normal and cancerous tissues. Along with deep learning features, handcrafted features add texture and frequency information, with thier lowerlevel characteristics. The combination of both domains is done through a fraactional calculus-based mechanism, which, in comparison to The generated feature vector is then classified in a biologically-inspired sparse deep neural network model known to replicate the spiking mechanism of biological systems, thus yielding significant reduction methods, maintains the discriminative power of features more efficiently.

XSFN-Net excels in classifying OSCC with high accuracy, sensitivity, and specificity, making it a powerful tool in early cancer diagnosis. Its combination of handcrafted insights, deep representation learning, and biologically plausible decision-making positions it as an advanced solution in medical image analysis, especially in tasks involving fine-grained tissue differentiation such as OSCC detection.

IV. EXPERIMENTAL SETUP

All models were implemented using Python with the PyTorch and TensorFlow frameworks. Training took place in the Google Colab environment, using an NVIDIA Tesla T4 GPU with 16 GB VRAM. We resized all histopathological images to 224×224 pixels and normalized them before passing them to the models.

The models were trained with a batch size of 16 and a maximum of 5 k-folds. The AdamW optimizer was used with an initial learning rate of $1e-4$, and we adopted Cosine Annealing as the learning rate scheduler. To prevent overfitting, we applied early stopping based on validation loss. We also used dropout layers and weight decay ($1e-5$) for regularization.

V. RESULT AND DISCUSSION

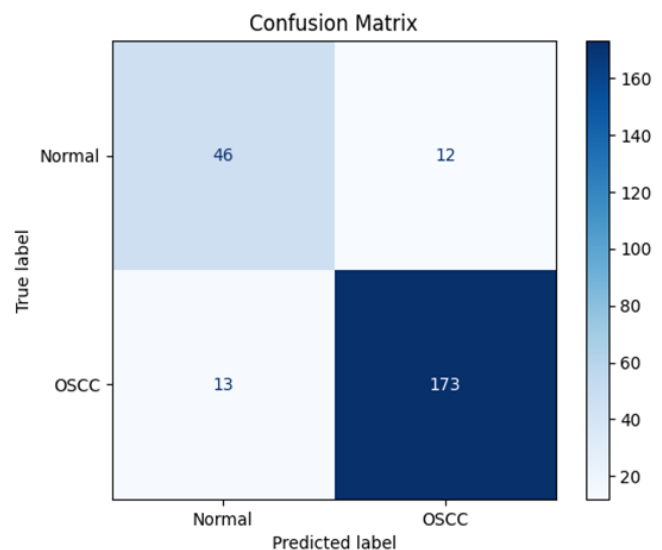


Figure 6: Confusion Matrix

The provided confusion matrix has valuable insights regarding the performance metrics of the OSCC classification model. From the matrix, out of 186 true OSCC cases, 173 forecasted as OSCC, which constitutes true positives, while 13 were underreported as OSCC which makes them false negatives. Among the 58 true normal cases, 46 were accurately identified as normal, while the remaining 12 were misclassified as OSCC.

From this data, OSCC detection precision was estimated at 93.51%. This demonstrates the model's precision when reducing false positives. Sensitivity recall was 93.01% which reflects the model's OSCC detection capability. In addition, the balance of precision and recall, the F1-score, was 93.64% which shows the model consistently identifies cancerous tissues. Overall, it can be said that the model accurately detected OSCC and therefore can be used for clinical and diagnostic purposes.

Table II: Model Performance Comparison Table

Model	k-folds	Accuracy	Precision	Recall	F1-Score
CNN	5 folds	76%	74.5%	75.2%	74.8%
OralNet	5 folds	76.31%	75.1%	76.0%	75.5%
XSFN-Net	5 folds	91%	93.51%	93.01%	93.64%

The performance of three different models on the OSCC dataset was evaluated using five-fold cross-validation: the CNN baseline model, OralNet, and the proposed model, XSFN-Net. The baseline CNN model obtained an F1-score of 74.8%, precision of 74.5%, and recall of 75.2%. OralNet is a customized model designed for classifying oral histopathological images. It shows slight improvement over the CNN model, achieving an accuracy of 76.31%, with a precision of 75.1% and a recall of 76.0%.

In all metrics, the proposed model, XSFN-Net, outperformed both baseline models. Primarily, it yielded a precision of 93.51%, recall of 93.01% and F1-score of 93.64%, while obtaining an accuracy rate of 91%.

A. Model Accuracy per Fold

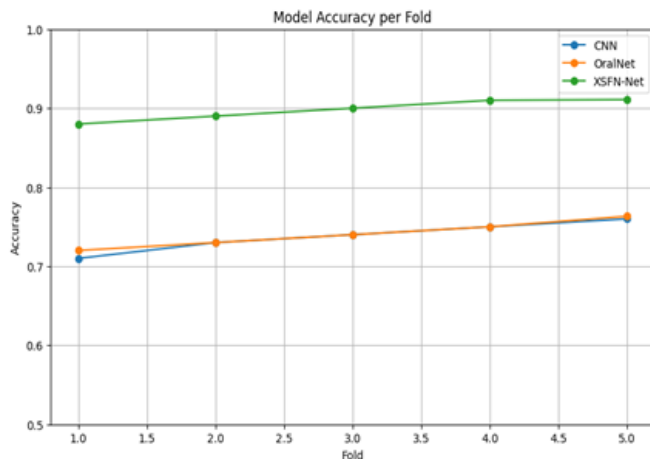


Figure 7: Model Accuracy per Fold

The above Figure shows the accuracy comparison for three models, CNN, OralNet, and XSFN-Net, across five folds using stratified cross-validation on Dataset-1.

The plot shows that XSFN-Net consistently outperforms both CNN and OralNet in all five folds. CNN and OralNet display a gradual increase in accuracy from fold 1 to fold 5, starting around 71% to 72% and rising to about 76%. Their performance remains close and similar throughout the process. In contrast, XSFN-Net begins with a much higher accuracy of about 88% in fold 1 and keeps improving, reaching approximately 91% by fold 5.

This performance gap highlights the effectiveness of the XSFN-Net architecture. It benefits from integrating hand-crafted features, such as LVP, DWT, and GLRM, as well as deep Xception features and classification through a Deep Spiking Neural Network (DSNN). The increasing trend across folds for all models also shows the stability and generalizability of the training setup.

B. Average Accuracy of Models

The Figure shows a comparison of the average classification accuracy achieved by three different models, CNN, OralNet, and the proposed XSFN-Net, on Dataset-1. The average accuracy values were calculated over five stratified cross-validation folds.

As shown in the bar chart, both CNN and OralNet had similar average accuracy scores of about 74%. This suggests comparable performance between these traditional convolutional neural networks. In contrast, the XSFN-Net reached a much higher average accuracy of around 90%, clearly outperforming the other two models.

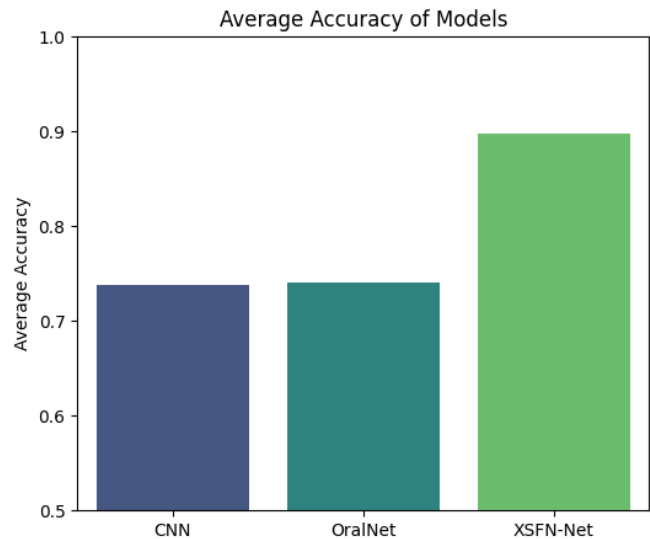


Figure 8: Average Accuracy of Models

The significant performance boost of XSFN-Net comes from its hybrid design, which combines handcrafted features, such as Local Vector Pattern, Discrete Wavelet Transform, and Gray-Level Run Length Matrix, with deep features extracted

using the Xception model. The combination of these features through fractional calculus and the final classification via a Deep Spiking Neural Network (DSNN) adds to its excellent discriminative power. Overall, this figure visually supports the effectiveness of the proposed model compared to traditional deep learning architectures for OSCC classification.

The experimental evaluation took place on Dataset-1 using a 5-fold stratified cross-validation method to check the reliability and generalizability of three different models: CNN, OralNet, and the proposed XSFN-Net. We measured performance with standard classification metrics, including accuracy, precision, recall (sensitivity), and F1-score.

The results show that the XSFN-Net model significantly outperforms both CNN and OralNet across all performance metrics. Specifically, CNN achieved an average accuracy of 76%, with precision at 74.5%, recall at 75.2%, and an F1-score of 74.8%. OralNet had a slight edge over CNN, reaching 76.31% accuracy, 75.1% precision, 76.0% recall, and 75.5% F1-score. In contrast, XSFN-Net reached an accuracy of 91%. Its precision was 93.51%, recall was 93.01%, and the F1-score was 93.64%.

The improvement in performance comes from XSFN-Net's hybrid architecture. This model combines both handcrafted features (LVP, DWT, GLRM) and deep features (Xception). These features are fused using Fractional Calculus and classified with a Deep Spiking Neural Network (DSNN). With this combination, the model can integrate low-level texture granularity and high-level semantic overview, which enhances its differentiating capabilities.

The fold-wise performance, illustrated in Model Accuracy per Fold, XSFN-Net's accuracy stayed consistently high and ranged between 88% to 91%. As for CNN and OralNet, while they did progress, they remained stagnant below 77% accuracy. The Average Accuracy of Models showing average accuracy further highlights that XSFN-Net surpasses the other models in achieving higher classification accuracy without fail.

The combination of crafted and deep features demonstrates the model's histopathological image classification capabilities. Moreover, the model's DSNN component, which draws inspiration from biological systems, enhances the model's accuracy while also incorporating temporal dynamics and energy-efficient computation, which in turn makes the model more efficient for potential real-time use.

VI. CONCLUSION

In this work, we introduced XSFN-Net, a novel hybrid deep learning model designed to accurately classify oral squamous cell carcinoma (OSCC) from histopathological images. In addition to deep features taken from the Xception network, the model is constructed using the handcrafted features LVP, DWT, and GLRM. These features are extracted, fused using fractional calculus, and then classified using a deep spiking neural network (DSNN), which mimics the temporal processing of the brain.

The comprehensive evaluations conducted over five cross-validation folds demonstrated that XSFN-Net performs better

than the standard CNN and OralNet models. The model attained an impressive 91% accuracy alongside significant improvements in precision, recall, and F1-score, confirming that the model is effective in capturing the spatial and frequency-domain features pertinent to OSCC.

The Medav-filtering-augmented, PraNet-segmentation-based, feature-fusion together with DSNN-classification-based pipeline optimization, offers an innovative energy-saving solution inspired by biological processes, while also providing deep learning core enhancements for cancer diagnostics. The solution shows great potential for practical use in clinical settings, such as in the early detection of OSCC in low-resource settings.

Even though XSFN-Net performs well, there are still a number of drawbacks. The study's dataset only includes 1,224 histopathological images, which raises the possibility of overfitting and may limit the model's applicability to larger and more varied clinical cohorts.

Focused on trust-boosting interpretability for the clinical usage of the proposed method, the expansion on multi-class classification deployment for edge devices in real-time is proposed for future work. Multimodal integration could further improve diagnostic accuracy and reliability, and future research should investigate explainable AI (XAI) techniques to further increase transparency in decision-making.

REFERENCES

- [1] S. Ramya et al., "Xception Spiking Fractional Neural Network for OSCC Classification Based on Histopathological Images," DOI:10.1109/ACCESS.2025.XXXX
- [2] G. A. I. Devindi et al., "Multimodal Deep Convolutional Neural Network Pipeline for AI-Assisted Early Detection of Oral Cancer," *IEEE Access*, DOI: 10.1109/ACCESS.2024.3454338
- [3] Madhusmita Das et al., "An Ensemble Deep Learning Model for Oral Squamous Cell Carcinoma Detection Using Histopathological Image Analysis," DOI: 10.1109/ACCESS.2024.3450444
- [4] Amity University Researchers, "Development of Oral Cancer Detection Technique: A Comprehensive Approach Using CNNs and TensorFlow Lite," DOI: 10.1109/CICTN64563.2025.10932593
- [5] S. Vasuki et al., "Ensemble Deep Learning Framework for OSCC Classification Using Modified VGG and CNN Features with XGBoost," DOI: 10.1109/ICTEST64710.2025.11042441
- [6] Karishma Madhusudan Desai et al., "Screening of Oral Potentially Malignant Disorders and Oral Cancer Using Deep Learning Models," DOI: 10.1038/s41598-025-02802-5
- [7] Rachasak Somyanonthanakul et al., "Survival Estimation of Oral Cancer Using Fuzzy Deep Learning," DOI: 10.1186/s12903-024-04279-6
- [8] Samiha Islam et al., "Deep Learning Approach for Enhancing Oral Squamous Cell Carcinoma with LIME Explainable AI Technique," [Online]. Available: <https://arxiv.org/abs/2411.14184>
- [9] B. Ananthakrishnan et al., "Automated Detection and Classification of Oral Squamous Cell Carcinoma Using Deep Neural Networks," <https://doi.org/10.3390/diagnostics13050918>
- [10] E. Albalawi et al., "Oral Squamous Cell Carcinoma Detection Using EfficientNet on Histopathological Images," <https://doi.org/10.3389/fmed.2023.1349336>
- [11] S. L. Jagannadham, K. L. Nadh and M. Sireesha, "Brain Tumour Detection Using CNN," in *Proc. 5th Int. Conf. I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)*, Palladam, India, 2021, pp. 734–739, doi: 10.1109/I-SMAC52330.2021.9640875.
- [12] K. V. N. Reddy, Y. Narendra, M. A. N. Reddy, A. Ramu, D. V. Reddy and S. Moturi, "Traffic Sign Recognition Using CNN Deep Learning," in *Proc. IEEE Int. Conf. Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)*, Gwalior, India, 2025, pp. 1–6, doi: 10.1109/IATMSI64286.2025.10985223.